RARE DISEASE, REAL KID · MVA HACKATHON 2026 · TRACK 1 METHODS REPORT

Biallelic *BUB1B* in a child with mosaic variegated aneuploidy

Three independent analysis arms converge on a null + hypomorphic allele pair (MVA type 1)

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Code: github.com/aperezcastro/hackaton-genetico (CC BY 4.0)

This report shows only the submitted candidate variants and aggregate statistics. No genome-scale table, read or genotype list leaves the protected compute environment (§13).

Answer at a glance

Gene	*BUB1B* (15q15.1; NM_001211.6 / ENST00000287598.11, MANE Select), compound heterozygous, phase inferred
Allele 1 · null	chr15:40209701 T>G · c.2210T>G · p.(Leu737*) · stop-gained, exon 17/23, NMD predicted · ClinVar P/LP · **Likely pathogenic (9 points), Pathogenic with ClinVar corroboration**
Allele 2 · hypomorph	chr15:40220612 T>G · c.3006T>G · p.(Asn1002Lys) · kinase-domain missense, last exon · ultra-rare · AlphaMissense 0.92 · **VUS (4–5 points)**
Disease	MVA type 1 (OMIM #257300), autosomal recessive
Independent check	Exomiser 15.1 ranks *BUB1B* first genome-wide, with and without a phenotype pre-filter
Submission	3 rows: the pair (EPCR 0.95) + two low-probability pairings with a third, deep-intronic *BUB1B* variant (0.03, 0.01)

4.96 M normalised variants in the VCF

1 gene left by the unbiased funnel and the panel

#1 *BUB1B* rank in Exomiser (of ~20,000 genes)

0 runs of homozygosity > 2 Mb (no autozygosity)

1 Summary

The genotype reproduces the textbook molecular architecture of MVA1. One allele is truncating (transcript degraded by NMD; no protein expected), and the other is a missense allele in the C-terminal kinase domain, where most MVA1 missense alleles act by lowering BUBR1 abundance (Hanks et al. 2004; Suijkerbuijk et al. 2010). Whether this particular allele lowers abundance or impairs function has not been measured; the Track 2 report carries both scenarios. The pair is architecturally analogous to the alleles modelled in mice by Sieben et al. (2020): BubR1 X753 (truncating) + L1002P (kinase-domain missense; mouse L1002 = human L1012). Our alleles are p.Leu737* and p.Asn1002Lys. That mouse genotype died prematurely, while the child is alive, which suggests the child's missense allele keeps more function.

Five design choices shaped the analysis:

(i) a **three-arm strategy** (Figure 1): an unbiased genome-wide search run in parallel with a hypothesis-driven MVA/spindle-assembly-checkpoint (SAC) panel and a phenotype-driven ranking, so that the answer is not just "where we looked";
(ii) because the second, hypomorphic MVA1 allele is often subtle, the panel was searched at **every consequence class** and scored for splicing;
(iii) **explicit, calibrated evidence**: ACMG/AMP criteria with Bayesian points for every submitted variant;
(iv) **genome-level checks from aggregate statistics**: allelic balance, mosaicism and autozygosity;
(v) **data protection by design**: patient data stayed in a private cloud environment, with no third-party

variant APIs.

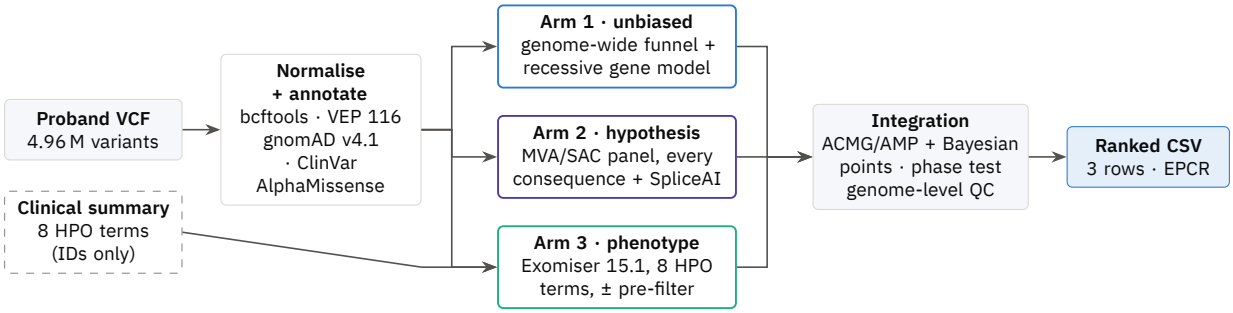


Figure 1. Analysis overview. Three arms (unbiased, hypothesis-driven, phenotype-driven) were run independently and only then integrated; all converge on the same *BUB1B* pair. Heavy steps run as time- and cost-capped jobs on Modal (total compute for Track 1 ≈ 1.4 USD, logged in `scripts/modal/gastos.csv`).

2 Data and reference

- **Input:** single-sample WGS VCF from the organisers (Sentieon 202308.02 + GATK VariantFiltration), GRCh38 no-alt analysis set + hs38d1 decoys, contigs without the `chr` prefix. No parental data, no BAM/CRAM.
- **Phenotype:** 8 HPO terms extracted from the clinical summary as IDs only and resolved locally against `hp.obo`: rhabdomyosarcoma (HP:0002859), nephrocalcinosis (HP:0000121), short stature (HP:0004322), failure to thrive (HP:0001508), skeletal muscle atrophy (HP:0003202), premature birth (HP:0001622), small for gestational age (HP:0001518), recurrent spontaneous abortion (HP:0200067, family history).
- **Coordinates:** GRCh38, 1-based, VCF-normalised (left-aligned, parsimonious), with the `chr` prefix required by the submission format.

Step	Tool / version	Script
Normalisation	bcftools norm `-m -any`, left-alignment against the Ensembl GRCh38 primary assembly	`anotar.py`
Annotation	Ensembl VEP 116 offline (`--everything`, MANE, HGVS); gnomAD v4.1 exomes + genomes; ClinVar (cache + NCBI VCF with review status); SIFT, PolyPhen-2, AlphaMissense (hg38)	`anotar.py`
Funnel + panel	Python, per chromosome on Modal	`filtra.py`
Splicing	SpliceAI 1.3.1, GENCODE grch38 annotation, windows of 50 and 2,000 nt	`splice_ai.py`
Phenotype ranking	Exomiser 15.1.0, data release 2512 (hg38, phenotype), hiPHIVE, REVEL/MVP/AlphaMissense/SpliceAI	`exomiser.py`
Phase feasibility	heterozygous-site spacing in the inter-allele interval	`fase_viable.py`
Genome-level QC	allelic balance per chromosome and in 5-Mb bins; runs of homozygosity	`balance_alelico.py`
Validation	organisers' `evaluation.py`	—

Table 1. Pipeline components (in `scripts/modal/`; the structural context of Figure 4 comes from `scripts/t2/estructura.py`). `evaluation.py` is the organisers' public scorer from the hackathon Space.

3 Arm 1 — unbiased genome-wide funnel

Thresholds. AF < 1 % is deliberately permissive for a recessive disease with fewer than 50 known patients. The maximum credible population AF of a pathogenic recessive allele is far lower, but a loose cut avoids discarding hypomorphic alleles that may be enriched in a founder population. 65,992 variants remain at AF < 0.1 %.

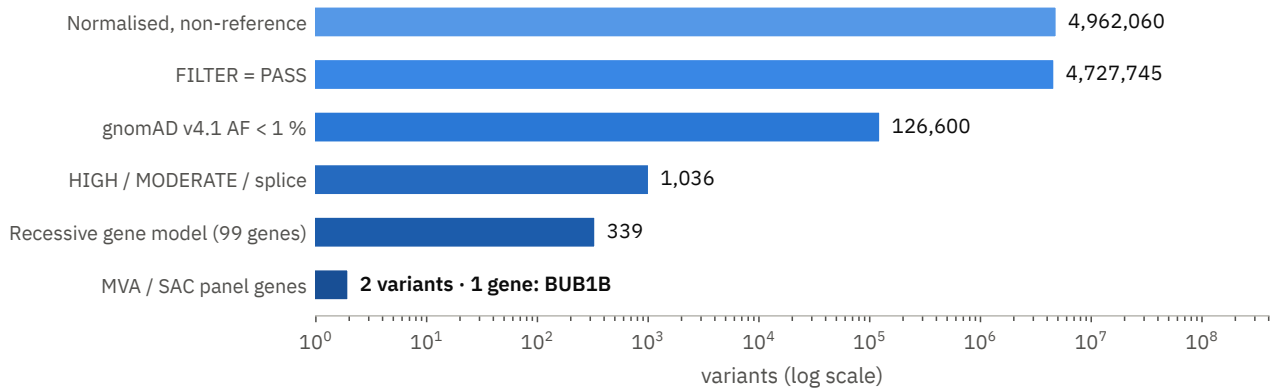


Figure 2. Variant funnel. Each step is a single, stated criterion. The recessive gene model keeps genes with ≥2 heterozygous or 1 homozygous qualifying variants; of its 99 genes only *BUB1B* belongs to the MVA/SAC panel. The other 98 genes were reviewed one by one (§9).

4 Arms 2 and 3 — the MVA/SAC panel and the phenotype

Panel (`filtra.py`): established MVA genes (Malumbres 2024) *BUB1B* (MVA1), *CEP57* (MVA2; Snape et al. 2011), *TRIP13* (MVA3; Yost et al. 2017) and *CENATAC/CCDC84* (MVA4); SAC and kinetochore genes (*BUB1, BUB3, MAD1L1, MAD2L1, MAD2L1BP, TTK, KNL1, ZW10, KNTC1, SPC24/25, CENPE, INCENP, BIRC5, SGO1*…); cohesion and SMC5/6 genes (*SMC5, SMC6, NSMCE2, SLF2*…); centrosome and minor-spliceosome genes. 313 variants of any consequence at AF < 5 %; 42 PASS at AF < 0.1 %.

Splicing. Of those 42, the 32 lying inside annotated gene bodies were scored with SpliceAI (Jaganathan et al. 2019), at two windows (50 and 2,000 nt) so that distant cryptic-site activation is not missed. The other 10 are 9 upstream/downstream variants and 1 intronic variant outside the SpliceAI annotation. The maximum delta in the panel is 0.03, far below the 0.2 threshold of a possible effect (Figure 3b). No splice-altering effect was predicted for any of the scored rare variants in the MVA/SAC genes.

Exomiser (Smedley et al. 2015), with the 8 HPO terms, all inheritance modes, PASS only and max AF 2 %, was run in two configurations: A, standard, with the hiPHIVE score ≥ 0.501 pre-filter; and B, without any phenotype pre-filter, to guard against confirmation bias. Both give the same top three (Figure 3a): *BUB1B* under a dominant model (the truncating allele alone) and under a recessive model (the pair; phenotype score 0.79), then *FANCD2* (two adjacent splice-region deletions on a non-MANE transcript, both ClinVar benign: excluded); ranks 4–10 are not named (§9). The gap between *BUB1B* (0.59 dominant, 0.56 recessive) and every other gene (≤ 0.22) shows that the answer is driven jointly by genotype and phenotype, not by either alone.

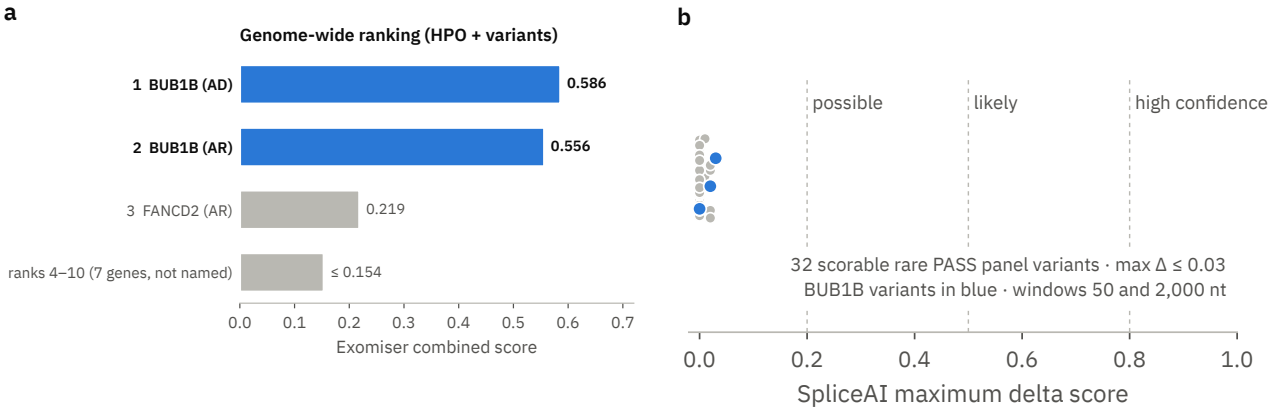


Figure 3. a, Exomiser configuration A (configuration B gives the same top 3). Only the genes discussed in the text are named; ranks 4–10 are shown only in aggregate. **b**, SpliceAI maximum delta for the 32 scorable rare PASS variants of the MVA/SAC panel (*BUB1B* variants in blue): none approaches the 0.2 threshold.

5 Candidate variants and ACMG/AMP classification

Criteria follow Richards et al. (2015) with ClinGen SVI refinements: the PVS1 decision tree (Abou Tayoun et al. 2018), PM2 downgraded to Supporting, the PM3 point system for recessive disorders, and splicing guidance for BP4. Points follow Tavtigian et al. (2020): Supporting 1, Moderate 2, Strong 4, Very strong 8, benign criteria negative; Pathogenic ≥ 10, Likely pathogenic 6–9, VUS 0–5 (Figure 5).

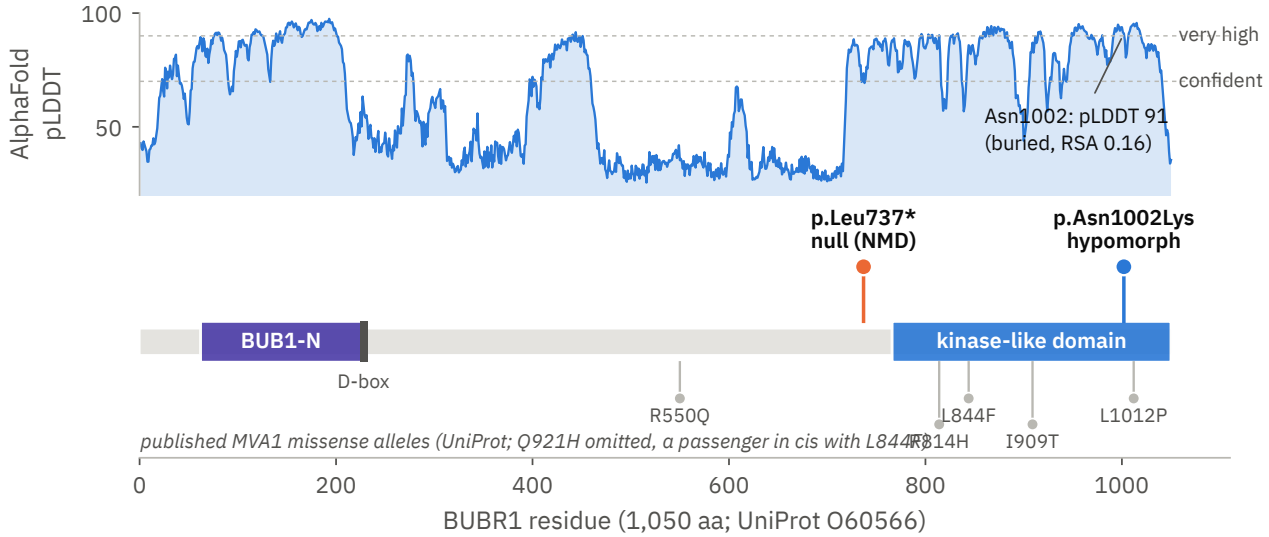


Figure 4. Where the alleles fall on BUBR1. Top, AlphaFold confidence (pLDDT) of model AF-O60566-F1 v6. Bottom, UniProt domains, the two alleles of this child (coloured) and the published MVA1 missense alleles (grey), 5 of 6 in the kinase-like domain. Asn1002 is a high-confidence, largely buried residue (relative accessibility 0.16) packing against residues 973–978 of the C-lobe, 12 Å from L1012. A longer, positively charged lysine there could destabilise the fold, as the kinase-domain MVA1 missense alleles tested do (Suijkerbuijk et al. 2010); the known stable-but-dysfunctional alleles lie outside this domain, but this residue has never been characterised, so the class of this allele is unmeasured.

5.1 Allele 1 — c.2210T>G p.(Leu737*)

The change converts the Leu codon TTA into a TGA (UGA) stop codon; BUBR1 is 1,050 amino acids long.

Criterion	Evidence	Points
PVS1	Nonsense in exon 17 of 23: premature stop > 50 nt upstream of the last exon–exon junction, so NMD is predicted. Loss of function is the established mechanism of MVA1 (truncating alleles are degraded by NMD and yield no detectable protein; Suijkerbuijk et al. 2010). Biologically relevant (MANE) transcript	+8
PM2_Supporting	gnomAD v4.1 AF 7.9×10^{-5} (exomes), 3.3×10^{-5} (genomes), no homozygotes; compatible with a recessive disease	+1
PM3	*not applied.* The ClinGen SVI point system requires the *trans* allele to be Pathogenic or Likely pathogenic; our partner allele is a VUS, and awarding PM3 here would make the two classifications bootstrap each other	0
(ClinVar)	Pathogenic / Likely pathogenic, criteria provided, multiple submitters, no conflicts. Corroboration only; PP5 was retired by ClinGen SVI	—
Total		**9 · Likely pathogenic**

Call quality: heterozygous, depth 46, allele fraction 0.54, GQ 99, PASS.

5.2 Allele 2 — c.3006T>G p.(Asn1002Lys)

Criterion	Evidence	Points
PM2_Supporting	ultra-rare: absent from the VEP-annotated gnomAD v4.1 release; at most one allele in the gnomAD v4 exome NFE data used by Exomiser	+1
PP3	AlphaMissense 0.923 (Cheng et al. 2023), MVP 0.852, PolyPhen-2 0.997, SIFT 0.01; REVEL 0.472 is indeterminate, so PP3 is applied at Supporting only	+1
PM1_Supporting	substitutions in or near the kinase-like domain (UniProt 766–1050) lower BUBR1 abundance 5–10×, the documented disease mechanism (Suijkerbuijk et al. 2010); 4 of the 5 MVA1 missense alleles annotated in UniProt sit there (Q921H is excluded: it is a functionally normal passenger in *cis* with L844F); structural context in Figure 4	+1
PM3_Supporting	with a pathogenic truncating allele in the same gene, phase unknown	+1
PP4 (contestable)	strictly PP4 asks for a *single* genetic aetiology and MVA has ≥4 genes, so a reviewer may drop it; we apply it at Supporting because the cytogenetic phenotype plus rhabdomyosarcoma, a tumour recurrently reported in MVA1	+1 (Moderate +2 defensible)
BP1 (proposed by Exomiser)	**not applied**: BP1 requires truncation to be the *only* mechanism; hypomorphic missense alleles are an established MVA1 mechanism	0
Total		**4–5 · VUS** (upper end if PP4 is accepted)

This allele stays a VUS on points, and we say so rather than claim a borderline Likely pathogenic: dropping the contestable PP4 leaves 4. Confirming *trans* phase would raise PM3 to Moderate and, with PP4, reach Likely pathogenic — which is why parental testing is the single most valuable experiment here. Call quality: heterozygous, depth 28, allele fraction 0.46, GQ 99, PASS. SpliceAI delta 0.02.

5.3 Alternative second allele — c.2679-1026A>G (deep intronic)

Absent from gnomAD; allele fraction 0.39. SpliceAI delta 0.00 at both windows gives BP4 (−1); with PM2_Supporting (+1) the total is 0, a VUS that leans benign. It is kept in the CSV only as a low-probability hedge (EPCR 0.03 and 0.01).

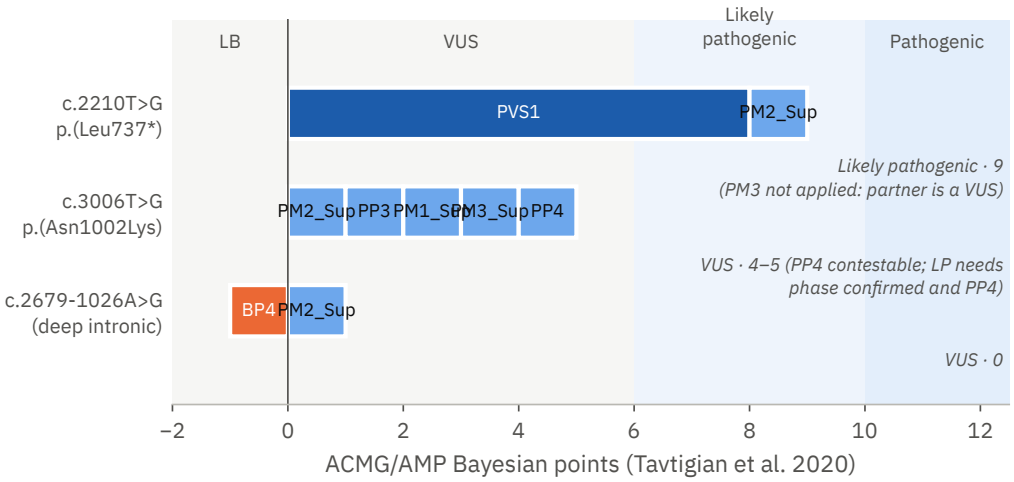


Figure 5. Bayesian ACMG/AMP points per submitted variant (Tavtigian et al. 2020). Shaded bands are the class thresholds.

6 Phase

The VCF carries no read-backed phase (`PGT`/`PID`) for these sites. The two alleles are 10,911 bp apart. With PCR-free Illumina libraries no read or read pair can span them, and phasing through intermediate heterozygous sites would need a chain of informative sites less than ~500 bp apart. We tested this

directly (`fase_viable.py`; Figure 6). The interval contains only three PASS heterozygous sites, separated by 6,769 and 4,142 bp, with no phase block, so no short-read chain can bridge them. Realigning the FASTQ (≈1 USD targeted, 5–8 USD genome-wide) therefore cannot phase the pair, and was not done.

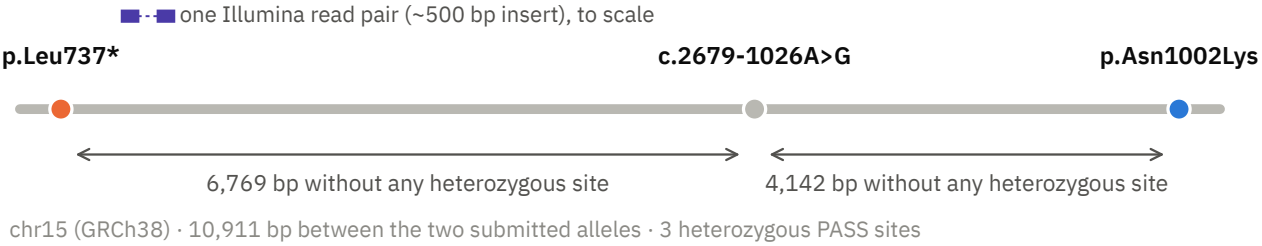


Figure 6. Why the pair cannot be phased from short reads: the only heterozygous sites in the interval are far apart compared with one read pair, drawn to scale.

Phase is argued from evidence instead:

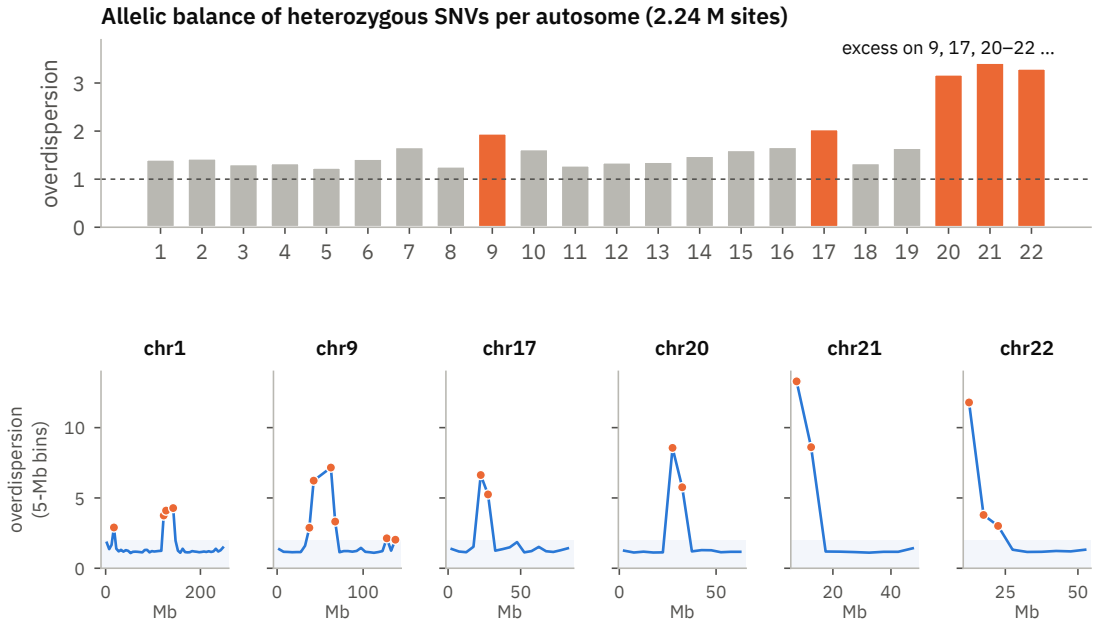
1. A *cis* configuration would leave one fully wild-type *BUB1B* copy. Monoallelic loss gives at most the milder premature chromatid separation trait (Matsuura et al. 2006), never full MVA; BUBR1 function is dosage dependent, with aneuploidy appearing below ∼50 % residual protein (Bohers et al. 2008).
2. The patient has a cytogenetic MVA diagnosis, and the genome has no other qualifying variant in any MVA/SAC gene, no second hit elsewhere in *BUB1B*, and no alternative recessive diagnosis (§9).
3. Both allele fractions are balanced (0.54 and 0.46). This is a quality control rather than a phase argument: it shows that both calls are genuine heterozygotes and excludes a large deletion of the other haplotype over either site; the 15q15 region is balanced as a whole (§7).

Trans is by far the most parsimonious configuration. Recommended clinical confirmation: targeted parental testing, or long-read sequencing of the 11-kb interval.

7 Genome-level checks: allelic balance, mosaicism and autozygosity

Three checks computed in the protected environment (`balance_alelico.py`). Only per-chromosome statistics leave it.

1. **Allelic balance and mosaicism.** Per-chromosome overdispersion is 1.24–1.49 on most autosomes but higher on 9, 17 and 20–22. Binning at 5 Mb shows that **the excess is focal**: it is confined to pericentromeric and segmental-duplication regions, where depth is inflated (61–105× vs 44×) and the allele fraction is depressed (0.30–0.40). That is the signature of collapsed paralogues, not of aneuploidy. Everywhere else every bin is balanced. No whole-chromosome or arm-level imbalance is detectable, which is expected in MVA: aneuploidy is *variegated* (different chromosomes in different cells, each at a few percent) and cannot be resolved by bulk allele fractions. Karyotyping or single-cell methods remain the diagnostic standard.
2. **The *BUB1B* locus.** *BUB1B* ± 5 Mb: 6,312 heterozygous sites with median allele fraction 0.50, so no loss of heterozygosity or imbalance around the gene. This supports both variants being genuine germline heterozygous calls.
3. **Autozygosity.** No run of homozygosity longer than 2 Mb in callable sequence. Runs were counted between consecutive heterozygous sites that still contain called homozygous-alternate sites, which excludes assembly gaps and centromeres; `bcftools roh` agrees. Extended autozygosity would have favoured a homozygous allele from consanguineous parents. Its absence fits the compound heterozygous model and the two different heterozygous alleles observed.



... is focal: confined to pericentromeric / segmental-duplication bins (orange), where depth is inflated and allele fraction depressed (collapsed paralogues). Everywhere else every bin is balanced: no detectable whole-chromosome or arm-level imbalance.

Figure 7. Allelic balance of 2.24 M heterozygous SNVs (PASS, depth $\geq$ 20). Overdispersion is the observed variance of allele fractions divided by the variance expected from binomial sampling at each site's depth; a mosaic gain or loss would raise it along a whole chromosome or arm. Dashed line: binomial expectation (1).

8 Mechanism: why this genotype explains the phenotype

BUBR1 is a core component of the mitotic checkpoint complex, which inhibits APC/C–CDC20 until every kinetochore is attached; it also stabilises kinetochore–microtubule attachments. Cells from patients with biallelic *BUB1B* mutations show an impaired checkpoint, misaligned chromosomes and low total BUBR1 (Suijkerbuijk et al. 2010). The truncating allele is removed by NMD, and the kinase-domain missense proteins tested are rapidly degraded; the known stable-but-dysfunctional alleles lie outside this domain. The residual BUBR1 is enough for viability but not for faithful segregation, hence mosaic aneuploidies in many tissues.

Phenotype (HPO)	Fit with biallelic *BUB1B*
Rhabdomyosarcoma	recurrently reported in MVA1 (Hanks et al. 2004; Nishitani-Isa et al. 2019); MVA is on the AACR list of childhood genomic-instability cancer predisposition syndromes (Nakano et al. 2024)
Growth failure, short stature, SGA	core MVA features
Skeletal muscle atrophy	early sarcopenia in BubR1-hypomorphic mice (Baker et al. 2004); sarcopenia correlated with mTORC1 hyperactivity in mice carrying a single MVA allele; the biallelic genotype closest to this child (X753/L1002P) died prematurely (Sieben et al. 2020)
Recurrent spontaneous abortion (family)	not explained by this genotype, which is viable in the child; not diagnostic
Nephrocalcinosis	not a classical MVA feature; left unexplained (possibly treatment- or prematurity-related; no nephrocalcinosis-gene variant passed the funnel)

9 Alternative hypotheses excluded

Hypothesis	What was checked	Result
Other MVA genes (*CEP57, TRIP13, CENATAC*) and the SAC panel	every consequence, AF < 5 %, SpliceAI	no rare qualifying or splice-altering variant

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Hypothesis	What was checked	Result
Other recessive diagnoses	the 98 non-panel genes of the funnel, one by one	hypervariable or paralogous regions (HLA, KIR, MUC, KRTAP, ZNF, olfactory/taste receptors, pseudogenes), ClinVar-benign variants, low-depth homozygous artefacts (2–6 reads), or no phenotype fit
Exomiser rank 3 (*FANCD2*)	variant-level review	two adjacent deletions on a non-canonical transcript, ClinVar benign: artefact
Dominant rhabdomyosarcoma predisposition	*TP53, DICER1, NF1*; RASopathy genes; Beckwith–Wiedemann (*CDKN1C*, H19/IGF2); *SMARCB1*; Lynch; *PTCH1/SUFU*	no rare functional variant
Other chromosome-instability syndromes (Bloom, Werner, Rothmund–Thomson, Roberts/*ESCO2*, Fanconi)	coding + splice	no biallelic qualifying variant
Mulibrey nanism (*TRIM37*), recently reported with MVA (PMID 41702694)	coding + splice + Exomiser	no rare functional variant; Exomiser score 0
Mosaic aneuploidy detectable in bulk data	allelic balance, 5-Mb bins	none beyond artefact regions (§7)
Incidental findings	genome-wide	handled per the organisers' IRB guidance; not reported

10 Ranking strategy and EPCR

Row	Variants	EPCR	Rationale
1	c.2210T>G + c.3006T>G	0.95	pathogenic null + hypomorph in the only gene supported by genotype and phenotype; Exomiser #1
2	c.2210T>G + c.2679-1026A>G	0.03	the only other rare *BUB1B* variant; SpliceAI 0.00, so very unlikely
3	c.3006T>G + c.2679-1026A>G	0.01	excludes the only pathogenic allele: least likely

EPCR values are a calibrated judgement. The residual 1–5 % covers phase uncertainty and the unobservable possibility of a structural or deep regulatory allele missed by short-read calls. The two hedge rows cost nothing if row 1 is correct, because F-max is taken at the best threshold. No secondary findings are submitted. The file was validated with the organisers' `evaluation.py`.

11 Limitations

- No BAM/CRAM: structural variants, copy-number changes and repeat expansions were not called genome-wide; the balanced allele fractions only exclude deletions spanning the two sites.
- Phase is inferred, not observed (§6).
- The VEP and Exomiser gnomAD snapshots differ slightly for allele 2 (absent vs one allele); both support PM2.
- Predictors disagree in magnitude for allele 2 (REVEL intermediate, AlphaMissense high), so PP3 was applied at Supporting only.
- Single proband: no segregation evidence (PP1).

12 Reproducibility

All scripts are in the repository; each Modal run is logged with its cost in `scripts/modal/gastos.csv`. Pinned versions: VEP 116 (cache 116, GRCh38), gnomAD v4.1, AlphaMissense hg38, ClinVar NCBI (October 2026), Exomiser 15.1.0 with data 2512, SpliceAI 1.3.1 (TensorFlow 2.15). Figures are regenerated by `scripts/figuras/figuras_t1.py` and this document by `scripts/compila_informes.py`.

```
python scripts/hpo_extraer.py                                          # HPO IDs (labels: local hp.obo)
python scripts/modal/refs.py && python scripts/modal/exomiser_refs.py              # public references
python scripts/modal/anotar.py --subir && python scripts/modal/filtra.py      # annotate, funnel, panel
python scripts/modal/splice_ai.py && python scripts/modal/exomiser.py             # splicing, phenotype
python scripts/modal/fase_viable.py                                                  # phasability test
python scripts/modal/balance_alelico.py [--diagnostico|--roh]                         # genome-level QC
```

13 Data protection and LLM use

- **LLM declaration** (organisers' format, discussion #2, 28 September 2026): *Anthropic Claude via the Claude Code CLI, Claude Max (consumer tier), model-training opt-out enabled ("Help improve Claude" off), no feedback provided on any output, no content flagged for safety review.*
- **Applicable data-handling terms** (discussion #21): opted-out conversations are not used for training and are retained for 30 days. The two exceptions (feedback, safety flags) were avoided, and feedback, bug-report and transcript-sharing commands were disabled by configuration.
- **Data minimisation**, as an added precaution: the model saw code, aggregate counts, HPO terms and short candidate lists ($\leq$ 50 annotated variants). The raw data and genome-scale derivatives in `data/` were blocked from the agent by configuration; intermediate tables were processed only by scripts, never loaded whole into the model context (`docs/llm_use.md`).
- **Compute:** Modal (data processor, no training on customer data). The case VCF lives in a private volume that will be deleted within 30 days of the close. No patient variant was sent to any third-party API; all annotation used downloaded public resources.
- **Incidental findings:** handled as advised by the organisers in discussion #20. They are not named in any public material, the family and clinical team are not contacted, and the organisers have escalated the question to the study IRB.

Acknowledgement

This work was made possible through the Hackathon, organized by Sage Bionetworks in partnership with the MVA Society, Hugging Face, and BEACON (The Benchmarking, Evaluation, and Assessment Consortium for Science), with prize sponsorship from AWS and Anthropic. We are deeply grateful to the child and their family who generously contributed their data and their story to advance research into this rare disease. We acknowledge their trust in making this Hackathon possible.

Dataset: SageBio. *Rare Disease, Real Kid: MVA Hackathon 2026 — Dataset*. Hugging Face, `SageBio/mva-hackathon-2026-data`, 2026.

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Yost S et al. Biallelic TRIP13 mutations predispose to Wilms tumor and chromosome missegregation. *Nat Genet* 2017. PMID 28553959.

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A Methods description form (Track 1 methods)

Answers to the organisers' official methods description template (version with the required LLM question), reproduced here because the submission form has no separate upload field for it.

Team name
aperezcastro (Agustín Pérez Castro, individual)

Model number (fill out this form once per model/approach submitted; up to 6)
1 (BUB1B compound heterozygous; aperezcastro_bub1b_comphet.csv)

Please describe your model/approach in detail.
Three-arm pipeline on the provided VCF, run as time-capped jobs on Modal (scripts in the GitHub repo). (1) bcftools norm (split multiallelics, left-align vs Ensembl GRCh38) → Ensembl VEP 116 offline (MANE, HGVS, gnomAD v4.1 exomes+genomes, ClinVar with review status, SIFT, PolyPhen-2, AlphaMissense). (2) Unbiased genome-wide funnel: PASS → max gnomAD AF < 1% → HIGH/MODERATE/splice → gene-level recessive model (≥2 het or 1 hom) = 339 variants in 99 genes, each reviewed. (3) Hypothesis-driven arm: MVA/spindle-assembly-checkpoint panel (BUB1B, CEP57, TRIP13, CENATAC, BUB1, BUB3, MAD1L1, MAD2L1BP, SMC5/6 complex…) at any consequence and AF < 5%; of the 42 rare PASS variants, the 32 inside annotated gene bodies were scored with SpliceAI 1.3.1 (50 and 2000 nt windows; all max delta ≤ 0.03). (4) Exomiser 15.1 (data 2512) with the 8 HPO terms, with and without a phenotype pre-filter: BUB1B ranks #1 genome-wide in both. (5) ACMG/AMP classification with ClinGen SVI refinements and Bayesian points (Tavtigian 2020) for each submitted variant; phase feasibility tested directly (only 3 heterozygous sites in the 10.9 kb interval, gaps 6.8 and 4.1 kb: not phasable with short reads) and argued from biology. (6) Genome-level QC from aggregate statistics: allelic balance per autosome and in 5-Mb bins (excess overdispersion confined to pericentromeric/segmental-duplication regions, i.e. technical; no detectable whole-chromosome imbalance, as expected for variegated low-level mosaicism), balanced allele fractions around BUB1B, and no run of homozygosity > 2 Mb (consistent with compound heterozygosity rather than autozygosity). Answer: BUB1B c.2210T>G p.(Leu737*) [Likely pathogenic, 9 points; Pathogenic with ClinVar corroboration] + c.3006T>G p.(Asn1002Lys) [VUS, 4-5 points], MVA type 1.

(required) If commercially-available Generative AI was used, record the provider, the plan or tier, and the relevant setting in your methods description. A line is enough, for example, "Anthropic API, Claude Sonnet, commercial terms, no training on customer content."
Anthropic Claude via the Claude Code CLI (Claude Opus models), Claude Max (consumer tier), model-training opt-out enabled ('Help improve Claude' off), no feedback provided on any output, no content flagged for safety review - per the organisers' confirmed guidance of 28 Sep 2026 (discussion #2). Applicable terms: opted-out conversations are not used for training and are retained 30 days. Additional data minimisation: the model only saw code, aggregate counts, HPO terms and short candidate lists (≤ 50 annotated variants); raw data and genome-scale derivatives were blocked from the agent by configuration. No patient variant was sent to any third-party API.

Please state whether the submission file is the automated output of your computational approach (preferred), or if it has undergone downstream manual review and curation.
Candidate generation is automated (funnel, panel, SpliceAI, Exomiser); the final 3-row file and EPCR values were curated manually from the automated evidence.

Please describe any downstream manual review in detail.
Manual review of the 99 recessive candidate genes (artefact-prone regions, ClinVar-benign, low depth, phenotype fit), of Exomiser's top hits, and of dominant rhabdomyosarcoma predisposition genes. Row 1 = the pair (EPCR 0.95). Rows 2-3 are low-probability alternative pairings with a third rare deep-intronic BUB1B variant (SpliceAI 0.00; EPCR 0.03 and 0.01). EPCR reflects residual uncertainty about phase and unobserved SV/regulatory alleles. Validated with the organisers' evaluation.py.

Please state whether your approach only used publicly available data (preferred) or if proprietary data was also used. If proprietary data were used, where possible, please also include a submission based only on publicly available data.
Only publicly available data.

Please describe any public data used in detail (e.g., reference panels, population databases, literature).
GRCh38 (Ensembl 116 primary assembly), Ensembl VEP 116 cache, GENCODE (SpliceAI grch38), gnomAD v4.1, ClinVar (NCBI VCF, Oct 2026), AlphaMissense hg38, Exomiser data release 2512 (REVEL, MVP, AlphaMissense, SpliceAI, HPO/MGI/ZFIN phenotypes), Human Phenotype Ontology (hp.obo), UniProt O60566 and the AlphaFold model, PubMed literature (cited in the report).

Please describe any proprietary data used in detail.
None.

Is your approach able to output proposed pairs of compound heterozygous candidate variants, or only single candidate variants?
Yes: pairs are produced by the gene-level recessive model (all qualifying pairs within a gene) and ranked; single variants are produced for dominant models.

How did you handle secondary or incidental findings, if any? (See the finding_type column in your submission.)
No secondary findings were submitted. Incidental findings were handled following the organisers' IRB guidance (discussion #20) and are not reported.

Please provide an estimate of run time and cost for your approach (if possible).
About 1 hour wall-clock of cloud compute (annotation parallelised over 25 containers ~20 min, Exomiser ~7 min, SpliceAI ~4 min) plus reference downloads; total Modal cost ~1.4 USD (scripts/modal/gastos.csv). Reproducible from the repository.

Provide a method abstract (up to 500 words). Please include strengths and limitations of your method as well as methodological detail.
We combined an unbiased genome-wide search with a hypothesis-driven analysis of genes linked to mosaic variegated aneuploidy (MVA). The single-sample WGS VCF was normalised and annotated offline with Ensembl VEP 116 (gnomAD v4.1, ClinVar, AlphaMissense) on a private cloud environment. A recessive gene-level funnel reduced 4.96 M variants to 339 variants in 99 genes, of which only BUB1B belongs to the MVA/spindle-checkpoint panel. Because the second, hypomorphic allele in MVA1 is often subtle, the panel was searched at every consequence class and scored with SpliceAI; independently, Exomiser 15.1 with the patient's HPO terms ranked BUB1B first genome-wide with and without a phenotype filter (score 0.59 dominant / 0.56 recessive vs $\leq$ 0.22 for the next gene). The proband carries BUB1B c.2210T>G p.(Leu737*) (nonsense, NMD predicted; ClinVar P/LP; ACMG 9 points, Likely pathogenic, Pathogenic with ClinVar corroboration) and c.3006T>G p.(Asn1002Lys) (ultra-rare kinase-domain missense, AlphaMissense 0.92, buried high-confidence residue in the AlphaFold model 12 Å from the known MVA1 allele L1012P; 4-5 points, VUS; its class, low abundance or impaired function, is unmeasured). This null + kinase-domain hypomorph architecture is the classical MVA1 pattern and mirrors a published mouse model (BubR1 X753 + L1002P, mouse L1002 = human L1012). A third rare deep-intronic BUB1B variant was assessed and deprioritised (SpliceAI 0.00). Phase could not be observed: the 10.9 kb interval contains only three heterozygous sites (gaps 6.8 and 4.1 kb), so we tested and documented that short-read phasing is infeasible and argued trans configuration from biology (monoallelic loss does not cause MVA), absence of any other qualifying variant and balanced allele fractions, and the genome shows no run of homozygosity > 2 Mb. Strengths: three independent arms converging on the same pair, explicit ACMG evidence with Bayesian points, sensitivity to subtle alleles, full reproducibility with cost-capped scripts, and strict data minimisation (no third-party variant APIs). Limitations: no BAM/CRAM, so structural variants were not called genome-wide; phase is inferred rather than observed; in silico predictors disagree in magnitude for the missense (REVEL 0.47); single proband without segregation data.